DATA SCIENCE PRACTITIONER-WORKPLACE MODULE

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Connecting data, people, and the cloud.

WM-02: Statistical Data Analysis Processes — Workplace Assignment

Learning outcomes:

- Clean and prepare workplace datasets including categorical enrichment (course/qualification).
- Apply descriptive statistics and visualize distributions across multiple dimensions (region, gender, course).
- Perform hypothesis tests (t-test, chi-square) and interpret results.
- Model outcomes using regression (linear, logistic) with categorical predictors.
- Communicate insights and recommend targeted interventions by course and career path.

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Workplace Assignment

Date: 16 Feb 2026

Module: 251102-001-00-WM-02 (Level 5, 16 Credits)

Dataset: *Training Centres Network Learner Performance Dataset (WM02_raw.csv)*

Context: Training centres dataset with learners, regions, attendance, exam scores, completion status, career paths, and course/qualification

1) Scenario & Purpose

You are a data analyst for a national training network. Management needs evidence to understand:

- How attendance and exam performance drive completion
- Differences across regions, genders, courses/qualifications, and career paths
- Whether the 60/60 rule (≥60% exam and ≥0.60 attendance) still makes sense for all contexts

Your analysis will inform funding, regional planning, course design, and targeted learner support.

2) Data You'll Use

- Fields: learner_id, gender, age, region, classes_attended, total_classes, exam_score, completion_status, career_path, course_qualification. (Expect messy values: missing exam scores, textual numbers like "eighty", attendance occasionally > total, and some missing genders. This is intentional.)
- File: WM02_raw_expanded_v4.csv (≈600 rows).

Note: You must validate the labelled completion against the 60/60 rule you compute from the cleaned data. Small inconsistencies were retained to test your approach.

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3) Tasks & Deliverables

A. Data Understanding & Cleaning

1. Audit missingness patterns, mixed types (e.g., textual exam scores), outliers (ages, attendance), and anomalies (attendance > total). Document the mechanism (MCAR/MAR/MNAR discussion) where relevant.
2. Propose two cleaning strategies; select one; justify each rule.
3. Engineer:
   - attendance_ratio = classes_attended / total_classes
   - age_bracket (16–20, 21–25, 26–30, 31+)
   - course_duration_weeks (extracted from the qualification text)
   - Validation: completion_by_rule (≥60 exam & ≥0.60 attendance) vs provided labels
4. Maintain a Data Quality Log (before/after snapshots; assumptions; residual risks).

Deliverable A: Data Quality Log (1–2 pages) + a short "Cleaning Decisions" summary (max ¾ page).

B. Descriptive Analytics

5. Summary stats for exam_score, attendance_ratio, age:
   - Overall and by region, gender, course, age bracket
   - Include 95% CIs and coefficients of variation where appropriate
6. Visualizations (clean, labeled):
   - Histogram of exam_score
   - Boxplots of exam_score by region and by course
   - Scatter: exam_score vs attendance_ratio with fit line
   - Bar chart: completion rate by course
   - (Optional for merits: heatmap of completion by region × course)

Deliverable B: Figures with captions (max 6–8 visuals) + 1 page narrative of key patterns.

C. Inferential Analytics

7. Welch's t test: Do mean exam scores differ between Completed vs Not Completed? Report Cohen's d and interpret magnitude.
8. Chi square tests:
   - completion_status × region
   - completion_status × course_qualification
   - Report Cramér's V; discuss practical significance.

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- (Optional distinction) Stratify by career_path and comment on stability.

Deliverable C: 1 page with tests, effect sizes, assumptions, and takeaways.

D. Predictive Modelling

9. Linear Regression:
    - Model: exam_score ~ attendance_ratio + age + gender (+ course_duration_weeks)
    - Report coefficients, CIs; check VIF, residual plots, and heteroskedasticity.
    - Compare a base model vs expanded model using AIC/BIC.
10. Logistic Regression (completion):
    - Model: completion_status ~ exam_score + attendance_ratio + course_qualification (+ region + age_bracket)
    - Use one hot encoding; present odds ratios with CIs.
    - Evaluate: confusion matrix, precision, recall, F1, ROC/AUC.
11. (Optional merit/distinction)
    - Add career_path interactions; 5-fold cross validation; compare to a decision tree.

Deliverable D: 2–3 pages of model results, diagnostics, metrics, and a short model selection rationale.

E. Recommendations & Executive Summary

12. Executive Summary (≤1 page): What drives completion? Which courses and regions need support? Are there gender/age nuances?
13. Actionable proposals (2–3): e.g., course specific academic support, threshold adjustments, targeted attendance interventions with monitoring plan.
14. Dashboard storyboard: List KPIs, visuals, filters, and user flow (wireframe acceptable).

Deliverable E: Executive Summary + Recommendations (2 pages) + storyboard (½–1 page).

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Category	Excellent:	Meets Standard	Developing	Below Standard
A. Data Understanding & Cleaning	**(20)** Thorough, consistent treatment of data issues with clear, justified decisions; changes are transparent and defensible.	**15:** Most issues handled; decisions generally clear.	**10:** Some issues addressed; limited clarity or consistency.	**5:** Major issues unaddressed or unclear.
B. Feature Engineering & Data Log	**10:** Helpful features created and validated; data decisions are clearly logged.	**8:** Key features present; basic logging.	**6:** Limited features and/or patchy logging.	**3:** Missing or incorrect features; no log.
C. Descriptives & Visuals	**20:** Accurate summaries with clear, professional visuals that illuminate patterns.	**15:** Correct summaries and readable visuals.	**10:** Partial summaries and/or unclear visuals.	**5:** Errors or visuals that do not support insight.
D. Inference	**15:** Appropriate tests with well-reasoned interpretation and practical meaning.	**12:** Correct tests with reasonable interpretation.	**8:** Partial, somewhat unclear, or weak interpretation.	**4:** Incorrect or unsupported conclusions.
E. Predictive Modelling	**20:** Well-specified models with appropriate checks; performance reported and interpreted.	**15:** Sound models with basic checks and metrics.	**10:** Models run but limited evaluation.	**5:** Models poorly specified or not interpretable.
F. Insights, Recommendations & Equity	**10:** Clear, actionable recommendations tied to evidence; considers equity/feasibility.	**8:** Actionable recommendations with some evidence.	**6:** Generic recommendations; weak links to evidence.	**3:** Vague or non-actionable recommendations.
G. Reproducibility & Appendix	**5:** Fully reproducible; figures/tables labelled; methods documented.	**4:** Mostly reproducible; minor gaps.	**2:** Some materials missing.	**1:** Not reproducible.

Notes: Your report should **show your thinking**, not just outputs. Use appropriate techniques to support your choices. Where trade-offs exist, **explain them**. Brevity + clarity = higher marks.

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